LAB 3: SIMULATION OF NETWROKING DEVICES USING CISCO PACKET

OBJECTIVE:

1. To study and understand the function of basic networking devices like: Hub, Switch, Bridge, Router and Repeater.
2. To understand the working principles and differences between basic networking devices
3. To understand the IP addresses used in connecting the different devices with each other.

THEORY:

Computer networks use various devices to connect computers and manage data transmission efficiently. Each device works at a different layer of the OSI model and performs a specific role in data communication.

1. Hub

A hub is a basic networking device that connects multiple computers in a network. It operates at the Physical Layer of the OSI model. When a hub receives data, it broadcasts the data to all connected devices, regardless of the destination.

Key Points:

- Works at Physical Layer
- Broadcasts data to all ports
- Less secure and inefficient

2. Switch

A switch is an intelligent networking device that operates at the Data Link Layer. It forwards data only to the intended destination device using MAC addresses.

Key Points:

- Works at Data Link Layer
- Uses MAC addresses
- Reduces network traffic
- More efficient than hub

3. Bridge

A bridge is used to divide a large network into smaller segments. It filters traffic based on MAC addresses and helps reduce collisions.

Key Points:

- Works at Data Link Layer
- Connects two network segments
- Reduces network congestion

4. Router

A router connects different networks and forwards data packets based on IP addresses. It operates at the Network Layer of the OSI model.

Key Points:

- Works at Network Layer
- Uses IP addresses
- Connects different networks
- Commonly used for internet access

5. Repeater

A repeater is used to regenerate and amplify weak signals in a network to extend the transmission distance.

Key Points:

- Works at Physical Layer
- Boosts signal strength
- Used for long-distance communication

IP Address in computer network

In this lab, IP addresses were used to uniquely identify each device in the network and enable communication between them. Without IP addresses, devices would not know where to send or receive data.

Importance of IP Address

- IP addresses provide a unique identity to each device on a network.
- They help devices locate each other and exchange data correctly.
- Routers use IP addresses to forward data packets between different networks.
- They prevent data from being sent to the wrong device.

IP Address ranges used in the lab:

192.168.1. x**:**

- Used for devices within the same Local Area Network (LAN).
- Devices with this IP range can communicate directly through switches or hubs.
- Example: PCs connected to the same switch using IPs like 192.168.1.2, 192.168.1.3, etc.

10.10.10.x**:**

- Used to represent a different network segment.
- Communication between 192.168.1.x and 10.10.10.x networks requires a router.
- Helps demonstrate inter-network communication.

OBSERVATION:

i. Hub:

PC Name	IP Address
PC-PT PC0	192.168.1.1
PC-PT PC1	192.168.1.2
PC-PT PC2	192.168.1.3
PC-PT PC3	192.168.1.4
PC-PT PC4	192.168.1.5
PC-PT PC5	192.168.1.6

ii. Bridge:

PC Name	IP Address
PC-PT PC19	192.168.1.1
PC-PT PC20	192.168.1.2
PC-PT PC21	192.168.1.3
PC-PT PC22	192.168.1.4
PC-PT PC23	192.168.1.5
PC-PT PC24	192.168.1.6

iii. Switch:

PC Name	IP Address
PC-PT PC13	192.168.1.1
PC-PT PC14	192.168.1.2
PC-PT PC15	192.168.1.3
PC-PT PC16	192.168.1.4
PC-PT PC17	192.168.1.5
PC-PT PC18	192.168.1.6

iv. Router:

Router: 1941 Router2	IP Address
GigabitEthernet0/0	192.168.1.1
GigabitEthernet0/1	10.10.10.1

PC Name	IP Address
PC-PT PC25	192.168.1.2
PC-PT PC26	192.168.1.3
PC-PT PC27	192.168.1.4
PC-PT PC28	10.10.10.2
PC-PT PC29	10.10.10.3
PC-PT PC30	10.10.10.4

v. Repeater:

PC Name	IP Address
PC-PT PC0	192.168.1.1
PC-PT PC1	192.168.1.2
PC-PT PC2	192.168.1.3
PC-PT PC3	192.168.1.4
PC-PT PC4	192.168.1.5
PC-PT PC5	192.168.1.6

DISCUSSION:

In this experiment, basic networking devices and IP addressing were studied. We were able to understand the roles of basic network devices in communication and transfer of data and information. Devices within the same IP range (192.168.1.x) communicated directly, while communication between different IP ranges (10.10.10.x) required a router. The experiment helped in understanding the role of networking devices and IP addresses in enabling proper data communication.

CONCLUSION:

Hence, the networking devices were simulated using Cisco packet.